

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: *Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)*

Assessment Tool Weightage: 10%

QUESTIONS: 4
Duration : 1 Hour

Total Marks:20

Technical Presentation

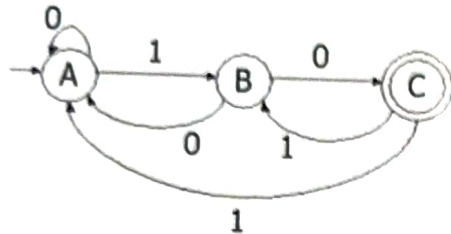
Code of Conduct:

I Deepak V-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

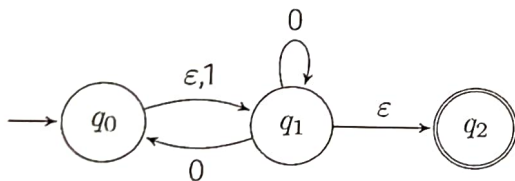
1. How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{0,1\}$. Construct a DFA that represents the same language L



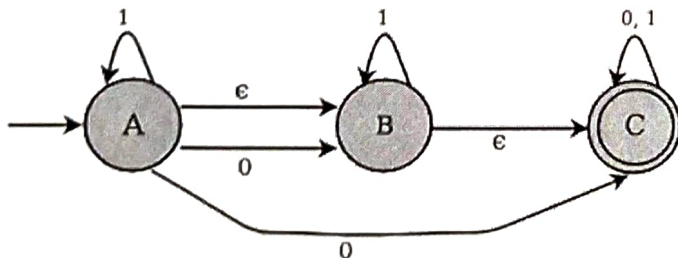
2. How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

3. What role do epsilon (ϵ) transitions play in finite automata, and how can ϵ -NFA be converted into an equivalent NFA without epsilon transitions? Construct an equivalent NFA without ϵ -transitions that accepts the same language. Draw its transition table and mark the initial and final states



4. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and construct an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states





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MATHEMATICAL INDUCTION AND ITS APPLICATIONS IN AUTOMATA THEORY

CSA1301–Theory Of Computation

Supervisor-
Dr.P. Vijayaragavan

Presented by-
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INTRODUCTION

- Mathematical induction proves statements for all positive integers.
- It is based on logical step-by-step reasoning.
- The method includes a base case and an induction step.
- It is widely used in mathematics and computer science.
- It ensures correctness of recursive formulas and algorithms.



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ROLE IN AUTOMATA THEORY

- Proves properties of recursively defined languages.
- Verifies correctness of automata and grammars.
- Establishes properties of strings of different lengths.
- Supports proofs of language operations and algorithms.
- Forms a foundation for formal verification.



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PROBLEM STATEMENT

- Prove the given sum of squares formula.
- Use the Principle of Mathematical Induction (PMI).
- Verify the statement for the base case.
- Assume the formula is true for $n = k$.
- Show that it is true for $n = k + 1$.



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BASE CASE

- Consider $n = 1$.
- Compute the left-hand side (LHS).
- Compute the right-hand side (RHS).
- Verify that LHS equals RHS.
- Therefore, the statement is true for $n = 1$.



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INDUCTION HYPOTHESIS

- Assume the statement is true for $n = k$.
- Use this assumption as the induction hypothesis.
- Substitute the hypothesis into the next step.
- Extend the proof to $n = k + 1$.
- Simplify the resulting expression.



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INDUCTION STEP

- Add $(k + 1)^2$ to both sides.
- Substitute the induction hypothesis.
- Factor and simplify the expression.
- Obtain the required formula for $k + 1$.
- Hence, the induction step is proved.



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CONCLUSION

- The base case is successfully verified.
- The induction step is successfully proved.
- The formula holds for all positive integers.
- Mathematical induction validates recursive statements.
- It is an essential proof technique in automata theory.

UBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

74/100